# Formula 1 Podium Prediction Companion

Milestone 1 analytical report · ACL Winter 2026 · Team 13: Youssef Hesham, Hatem Soliman, Mohammed Elazab, Tony Makaryous
Data: Kaggle *Formula 1 World Championship (1950-2024)*. Target: does a driver finish on the podium (`positionOrder <= 3`), predicted after qualifying and before the start.

**Summary**

- **Data.** 26,759 raw result rows become **24,921 rows, one per driver per race**, after removing the Indianapolis 500 (405 rows), cars that failed to qualify (1,368) and shared-drive duplicates (65). The podium rate goes from 12.7% to 13.4%.
- **Q1.** Front-row qualifiers reached the podium 65% of the time before 2014 and 74% since; the circuit ranking barely carries over between eras (rho = 0.22 over 13 circuits). Where the car actually starts matters more than where it qualified (29% vs 70% conversion).
- **Q2.** No home advantage once the grid slot is controlled (Mantel-Haenszel odds ratio 0.93, 95% CI 0.78-1.11).
- **Q3.** Mechanical retirements fell from 9.2% to 5.7% of starts. Highest rates: Lotus F1, Toro Rosso and McLaren in 2014-2021; Alfa Romeo and Alpine in 2022-2024.
- **Model.** 21 leakage-safe features, a chronological split (train up to 2019, validation 2020-21, test 2022-24) and a training window of 2003-2019 chosen on validation. The primary model, **gradient boosting** (highest validation PR-AUC, 0.734), reaches a **test PR-AUC of 0.711**, against 0.647 for the no-learning rule "top three on the grid". All learned models land between 0.69 and 0.72 on test; the differences between them are within noise, and the logistic-regression baseline does as well as anything (0.720).
- **Explanations.** The grid slot dominates, followed by the team's recent results, the gap to pole and the championship standings. These describe what the model relies on, **not what causes podiums**.

## 1. Data audit, cleaning and EDA

All 14 tables were read as plain text, so nothing was converted before we could measure it. Every step below is done on copies; the raw tables stay untouched.

- **Sentinels.** The literal `\N` appears in 30 of 120 columns. It is structural, not random: `position` is `\N` for every unclassified driver, Q2/Q3 only exist since 2006, and `qualifying.csv` itself only starts in 1994 and covers every race from 2003 (Figure 1). We turn `\N` and blank strings into real missing values and do not drop rows because of them.
- **Semantic types.** Ids and counters become nullable integers, dates become datetimes, and lap-time strings (`"1:27.452"`) get a numeric copy in seconds. The parsed lap and pit-stop durations match the separate `milliseconds` columns to within 10^-9 ms.
- **Keys.** All 14 primary keys are unique and non-null and all 23 foreign keys reconcile, before and after typing.
- **Scope.** We keep seasons up to 2024, remove the Indianapolis 500 (1950-60, a separate field of oval specialists) and remove the 1,368 cars that failed to qualify (`positionText = 'F'`), since they never started.
- **Grain.** 62 (raceId, driverId) pairs still had two or three rows, all shared drives from 1950-1964. Rule: keep the car the driver started in (`grid > 0`), then the lowest `resultId`. We did not keep "the best finish", which would pick rows using the label; that alternative would have changed 15 podium labels.
- `grid = 0` mostly marks cars without a grid slot (98.5% of non-qualifiers, 60% of withdrawn cars) and only occasionally a pit-lane start, so the model gets a `grid_zero` flag and those cars are placed behind the field.
- **Effect of cleaning** (Figure 5): fewer entrants per race in the 1980s-90s (no more non-qualifiers) and a correspondingly higher podium rate per season.

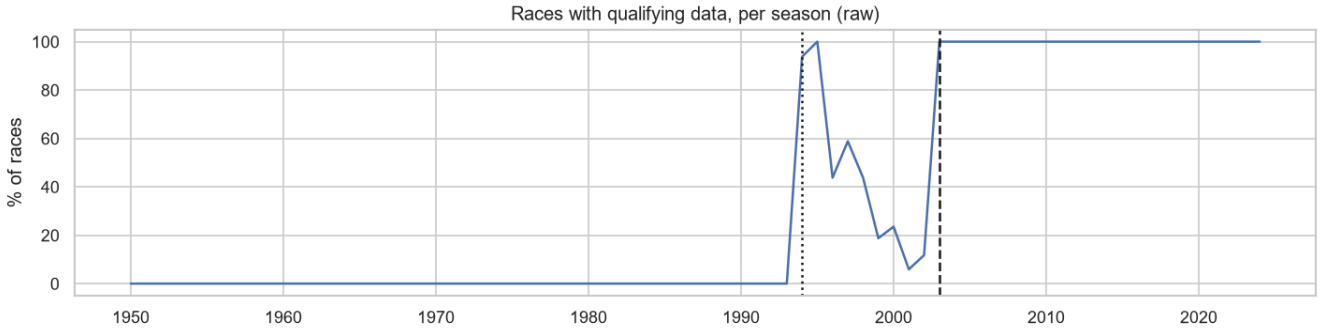



Figure 1. Races with qualifying data per season (raw files).

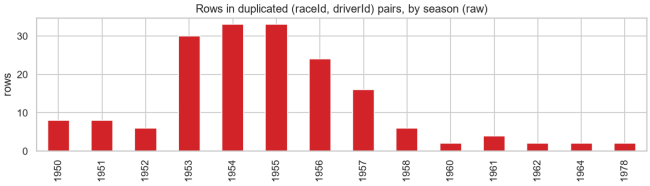



Figure 2. Rows in duplicated (raceId, driverId) pairs: shared drives in the 1950s-60s.

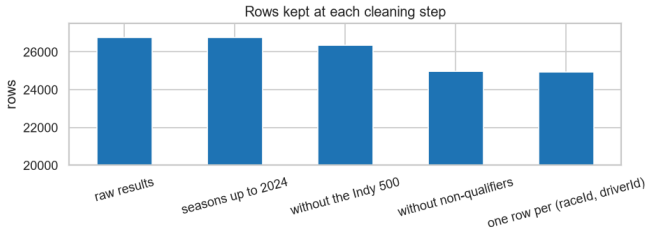



Figure 3. Rows kept at each cleaning step.

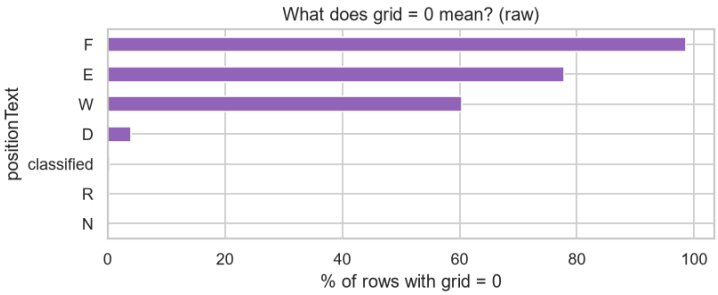



Figure 4. Share of rows with grid = 0 by finish code.

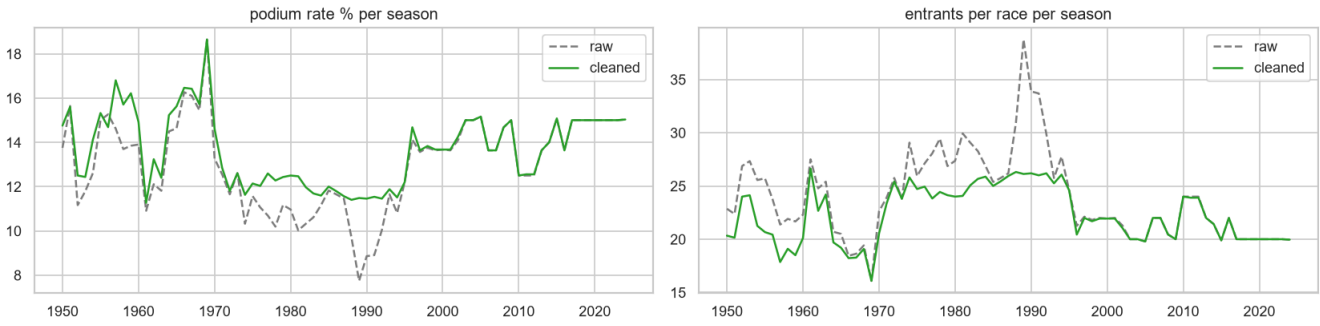



Figure 5. Podium rate and entrants per race, raw vs cleaned.

## 2. Data-engineering questions

### Q1. Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Join path `results -> races -> circuits` and `results -> qualifying` on (raceId, driverId); one row per driver per race. *Front row by qualifying* is a qualifying position of 1 or 2, *front row by grid* a grid slot of 1 or 2. Because qualifying only exists from 1994, both are computed on the same rows (races with qualifying data), and a circuit is ranked only with at least 12 front-row starts in each era. Both inputs are known before the start.

**Q1 answer.**

- **Best converters.** Since 2014 a front-row qualifier finished on the podium most often at **Suzuka (89%)**, **Monaco (85%)**, and **Spa and Barcelona (82% each)**. Before 2014 the top three were Suzuka (77%), Shanghai (70%) and Bahrain (67%).
- **Converting got easier overall.** Front-row qualifiers made the podium 65% of the time before 2014 and 74% from 2014 (by grid: 65% -> 76%). 11 of the 13 ranked circuits convert more often in the hybrid era; Bahrain stays level and the Hungaroring is the only one that drops (65% -> 59%).
- **Yes, the ranking changes a lot.** The rank correlation between the two eras is only rho = 0.22 (p = 0.48) using qualifying and 0.23 (p = 0.45) using the grid, which is statistically indistinguishable from "no relationship". Suzuka is first in both eras, but Spa jumps from #12 to #3, Barcelona from #8 to #3 and Monaco from #7 to #2, while Bahrain falls from #3 to #11 and the Hungaroring from #5 to #13.
- **Qualifying is not the same as the grid.** Drivers who qualified on the front row *and* started there reached the podium 70% of the time (n = 950). Those who qualified on the front row but started further back because of a penalty or a pit-lane start made it only 29% of the time (n = 38), while drivers promoted onto the front row by other people's penalties converted 64% (n = 36). Where the car actually starts matters more than the lap time that earned the slot.
- **Caveats.** Only 13 circuits have enough front-row starts in both eras, each with 6-17 races per era. Track layouts changed, and the post-2014 era was dominated by one or two teams, so circuit effects and team dominance are mixed together.

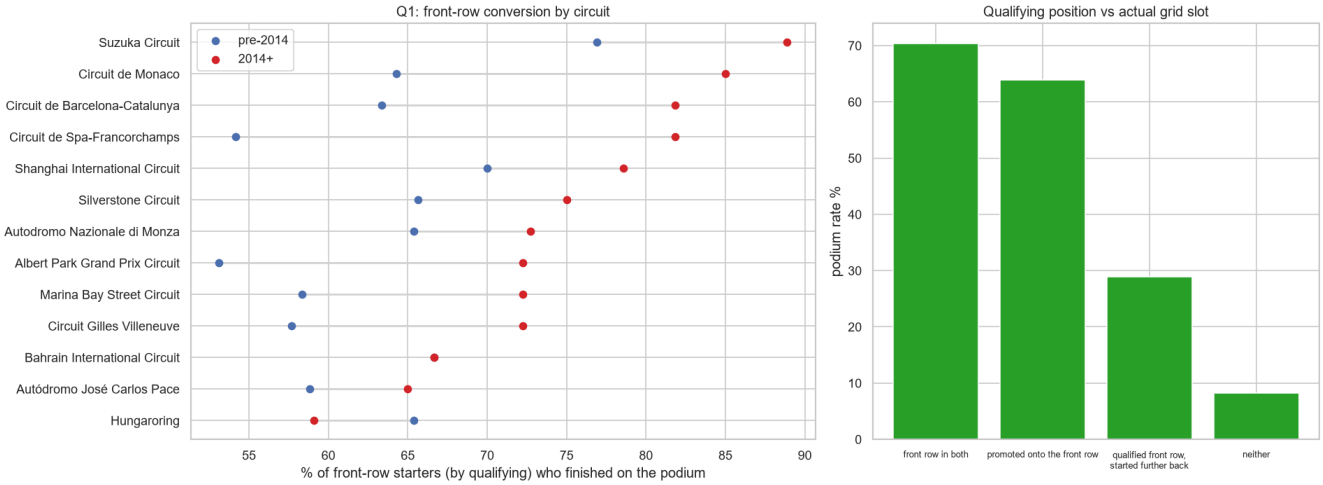



Figure 6. Left: front-row conversion per circuit, pre-2014 vs 2014+ (front row by qualifying). Right: qualifying position vs actual grid slot.

## Q2. Do drivers podium more often at home, after controlling for starting position?

Join path `results -> drivers.nationality` and `results -> races -> circuits.country`, with a hand-written nationality-to-country map. Rows with `grid = 0` are left out because they have no comparable slot. We compare home and away *within* grid buckets (Mantel-Haenszel) and with a logistic regression that adds era effects and clusters errors by driver.

**Q2 answer: no. Once we compare drivers who started from similar grid slots, there is no evidence that they podium more often at home.**

- **The raw numbers point the other way.** Home starters reached the podium 11.6% of the time against 13.7% away (1,941 home starts). That is mostly about *where* home drivers start: 15% of home starts are from P21 or lower, against 11% of away starts.
- **Within the same grid bucket the two curves lie on top of each other** (right panel). The pooled Mantel-Haenszel odds ratio is **0.93** (95% CI 0.78-1.11, p = 0.43), and the logistic regression with grid bucket and era gives **0.93** (95% CI 0.76-1.14, p = 0.50). Both intervals include 1.
- **What we can and cannot say.** A large home advantage is ruled out (the upper limits are +11% and +14% in odds), a small one is not. This is observational: driver quality is not fully captured by the grid slot, and the home sample is dominated by British, Italian, French and German drivers.

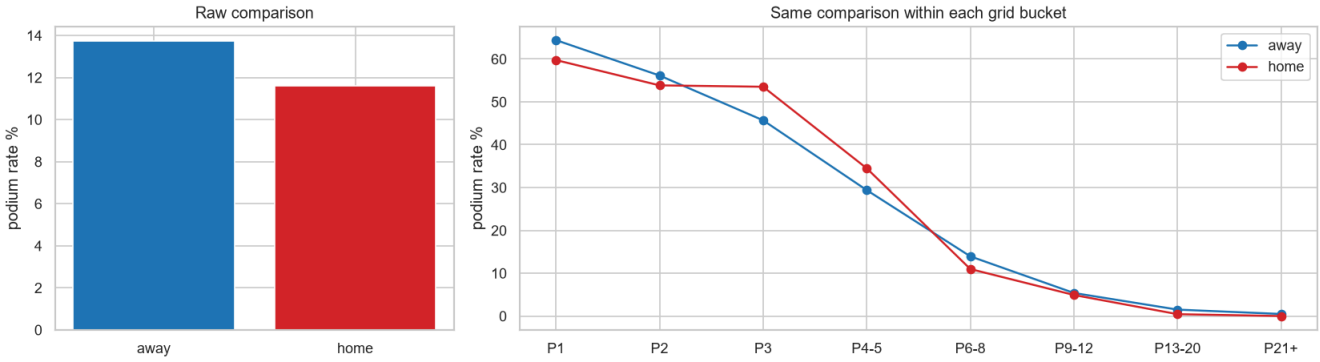



Figure 7. Podium rate home vs away: raw, and within grid buckets.

### Q3. Which constructors have the highest mechanical-retirement rate in 2014-2021 vs 2022-2024?

A mechanical retirement is a start that ended with a component-failure status (62 statuses such as engine, gearbox, hydraulics, power unit, brakes). Crashes, driver errors, regulation decisions and ambiguous statuses (puncture, damage, the generic "Retired") are not counted. The denominator is race starts, so withdrawn cars and non-qualifiers are excluded. `status` is post-race information and is used here only as a description of history, never as a model feature.

**Q3 answer.**

- **2014-2021** (3,258 starts, 9.2% of all starts ended in a mechanical retirement): the highest rates belong to **Lotus F1 (17.3%, 13 of 75)**, **Toro Rosso (16.1%, 39 of 242)**, **McLaren (13.8%, 44 of 318)**, Renault (13.5%) and Sauber (11.0%). Caterham was even higher (8 of 34) but has fewer than 40 starts, so it is not ranked.
- **2022-2024** (1,354 starts, 5.7%): **Alfa Romeo (12.5%, 11 of 88)** and **Alpine (11.0%, 15 of 136)** are clearly ahead of everyone else; next come AlphaTauri (6.8%), Ferrari (5.9%) and Williams (5.2%).
- **Reliability improved across the grid**, by 3.5 percentage points overall. The biggest improvements are at McLaren (13.8% -> 3.7%) and Red Bull (10.7% -> 3.7%). The Renault/Alpine works team is near the top in both periods (13.5% -> 11.0%).
- **Caveats.** In 2022-2024 a team has only 88-136 starts, so one or two failures move its rate by 1-2 points. Team renames (Toro Rosso -> AlphaTauri -> RB, Sauber -> Alfa Romeo -> Sauber, Renault -> Alpine) are kept under the name used in the dataset. 50 retirements are only recorded as "Retired" and could hide more mechanical failures.

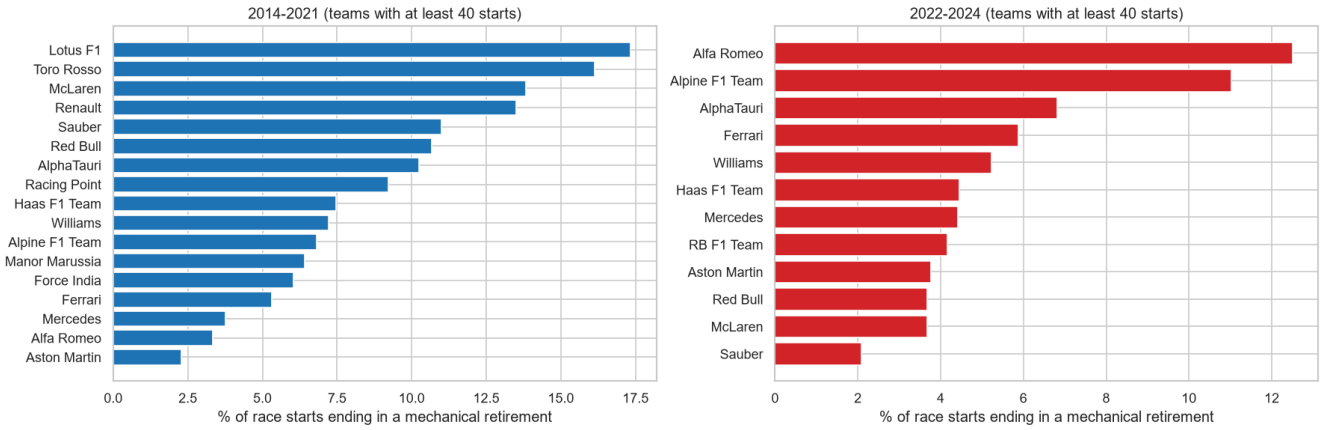



Figure 8. Mechanical-retirement rate per constructor and period (teams with at least 40 starts).

## 3. Features: what we use and why it is valid before the race

**Cutoff:** after qualifying, before the race. A feature is allowed only if it comes from

1. the qualifying session or the starting grid of *this* race, or

2. results and standings of **strictly earlier** races.

| Source | Allowed? |
|---|---|
| this race's `positionOrder`, `position`, `points`, `laps`, `time`, `fastestLap`, `rank`, `status` | no, these are the outcome |
| `lap_times`, `pit_stops` of this race | no, they happen during the race |
| `driver_standings` / `constructor_standings` of this race | no, they already include this race; we use the standing after the previous round instead |
| `sprint_results` | left out: only ~20 weekends since 2021, and the sprint result is already reflected in the grid |
| qualifying and `grid` of this race | yes |
| earlier races (form, standings, history at the circuit) | yes |

Races are put in chronological order with `(year, round)`. All "form" features are computed with `shift(1)`, so the current race is never included. We build a few more candidates than we need and choose between them in Section 3.

Team form is first averaged per team and race, so a team-mate's result in the same race cannot leak in, and standings come from the most recent *earlier* round of the same season. We checked this by recomputing three features from scratch for 200 random rows (0 mismatches in 600 values). A deliberately leaky version of the driver-form feature pushes validation PR-AUC from 0.705 to 0.831, which shows how much an undetected leak would have flattered the results.

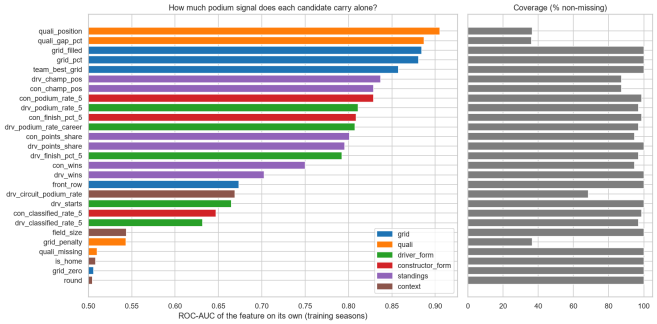



Figure 9. Signal of each candidate on its own (training seasons).

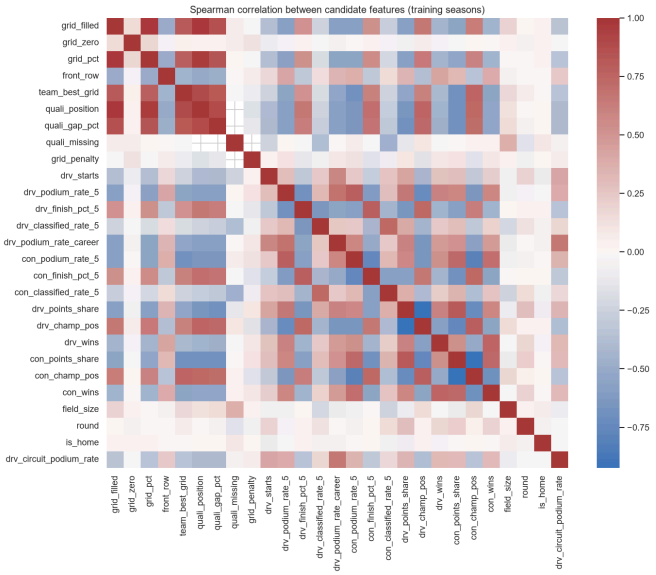



Figure 10. Spearman correlation between candidates.

**Decision.** The evidence above leads us to drop six candidates:

| Dropped | Why |
|---|---|
| `grid_pct` | rho = 0.971 with `grid_filled`: the same ordering divided by the field size |
| `quali_position` | rho = 0.975 with `grid_filled`, and missing for 63% of training rows. The *difference* between qualifying and grid, which Q1 showed matters, is kept as `grid_penalty` |
| `drv_champ_pos`, `con_champ_pos` | rho = -0.92 with the matching points share: the same information upside down |
| `round` | AUC 0.505 on its own, i.e. no signal |
| `is_home` | AUC 0.508 on its own, and Q2 found no home effect once the grid is controlled |

Everything we keep either carries signal on its own (AUC 0.54 to 0.89) or is needed to read another feature correctly: `grid_zero` tells the model that `grid_filled` was filled in, and `quali_missing` tells it that the

qualifying features were imputed. The binned plots show the shapes we expected: the podium rate falls steeply with the grid slot and the gap to pole, and rises with recent driver/team form and the points share. `grid_penalty` is the exception: drivers who start exactly where they qualified podium most often (about 16%), while both promoted and demoted drivers podium less (about 5%), so its effect is not a straight line and depends on where on the grid the change happens.

**The 21 features we keep, and why each one is known before the start:**

| Feature | Group | Meaning | Known before the start because |
|---|---|---|---|
| `grid_filled` | grid | grid slot; 0 (no slot / pit lane) becomes field size + 1 | the grid is final before the start |
| `grid_zero` | grid | 1 if the official grid value was 0 | same |
| `front_row` | grid | 1 if starting P1 or P2 | same |
| `team_best_grid` | grid | best grid slot of the team's cars in this race (a car-pace proxy) | both cars' slots are on the grid |
| `quali_gap_pct` | quali | % slower than the fastest qualifying lap of the race (capped at 30) | qualifying is over |
| `quali_missing` | quali | 1 if there is no qualifying row (mostly before 1994) | describes the data, not the race |
| `grid_penalty` | quali | grid slot minus qualifying position (> 0 = started further back) | both are known |
| `drv_starts` | driver_form | number of earlier starts | earlier races only |
| `drv_podium_rate_5` | driver_form | podium share in the last 5 starts | earlier races only (`shift(1)`) |
| `drv_finish_pct_5` | driver_form | average finishing position / field size, last 5 starts | earlier races only |
| `drv_classified_rate_5` | driver_form | share of the last 5 starts that were classified (reliability) | earlier races only |
| `drv_podium_rate_career` | driver_form | podium share over all earlier starts | earlier races only |
| `con_podium_rate_5` | constructor_form | team podium share over its last 5 races | earlier races only, team aggregated per race first |
| `con_finish_pct_5` | constructor_form | team average finishing position / field size, last 5 races | earlier races only |
| `con_classified_rate_5` | constructor_form | team share of classified cars, last 5 races | earlier races only |
| `drv_points_share` | standings | driver points / leader's points after the latest earlier round of the season | standings of earlier rounds only |
| `drv_wins` | standings | driver wins so far this season | same |
| `con_points_share` | standings | constructor points / leader's points (from 1958) | same |
| `con_wins` | standings | constructor wins so far this season (from 1958) | same |
| `field_size` | context | number of starters in this race | the starting list is fixed after qualifying |
| `drv_circuit_podium_rate` | context | driver's podium share at this circuit in earlier visits | earlier races only |

**Reading the trials** (validation PR-AUC, three model families with fixed settings):

- Starting-order information does most of the work. The qualifying group alone reaches 0.62-0.63, adding the grid group brings all three models to 0.66-0.67, and the championship standings add the next big step (0.715-0.729).
- After that the groups help some models and not others. Constructor form lifts the neural net to 0.735 but not the other two; driver form and context move the three models in different directions (between -0.03

and +0.01). Section 6 measures every group again after tuning.

- **Dropping the six features costs nothing measurable:** the mean over the three models is 0.7181 with the selected 21 features and 0.7208 with all 27 candidates. That 0.003 is smaller than what we get just by changing a random seed (Section 6), so we keep the smaller, less redundant set.

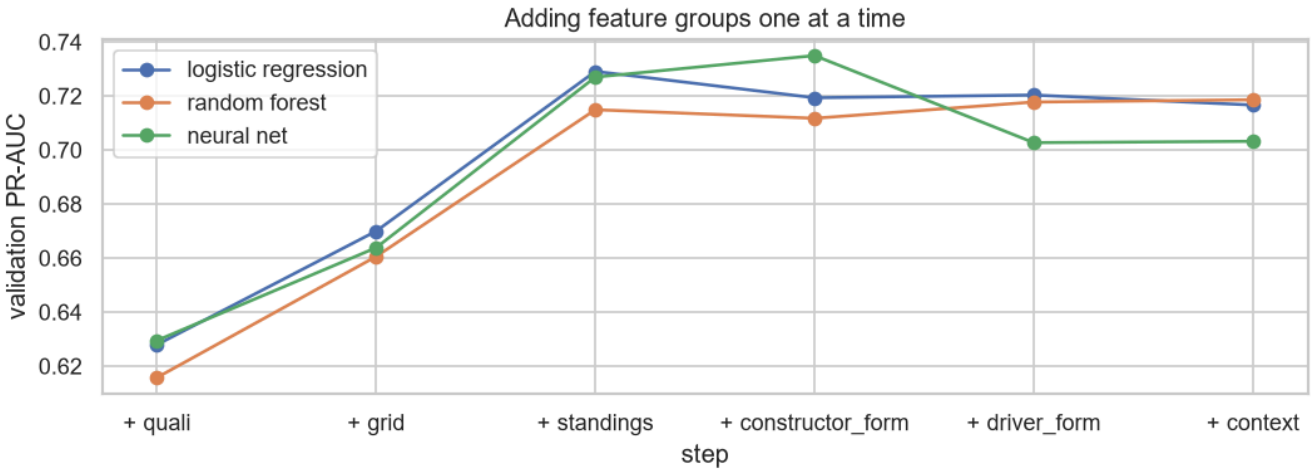



Figure 11. Validation PR-AUC while adding feature groups one at a time.

## 4. Split and preprocessing

**Split.** Train 1950-2019 (22,782 rows), validation 2020-2021 (39 races, 780 rows), test 2022-2024 (68 races, 1,359 rows). The split is chronological because form and standings are built from earlier races and because the real use is "predict the next race from the past". 2022-24 is a new regulation era, the toughest unseen test. Validation is used for every choice (hyperparameters, thresholds, training window); the test seasons only for the final numbers.

**Preprocessing.** Median imputation with missing-indicator columns, then standard scaling, fitted on the training rows only and shared by all models. Each experiment below changes one thing and is scored with plain logistic regression:

| variant (logistic regression) | val PR-AUC | val ROC-AUC | test PR-AUC | test ROC-AUC |
|---|---|---|---|---|
| reference | 0.7166 | 0.9304 | 0.7193 | 0.9377 |
| E2 scale -> impute | 0.7165 | 0.9301 | 0.7195 | 0.9377 |
| E3a no missing indicators | 0.7195 | 0.9281 | 0.7186 | 0.9374 |
| E3b no scaling | 0.7181 | 0.9305 | 0.7178 | 0.9375 |
| E1 fit preprocessing on all seasons -> split | 0.7150 | 0.9303 | 0.7198 | 0.9378 |

**What E1-E3 show.**

- **E1, fit order.** Fitting the imputer and scaler on every season before splitting changes validation PR-AUC by only -0.002 (0.7150 vs 0.7166). Adding five seasons to seventy hardly moves a median or a standard deviation, so the leak is small here. We still fit on the training rows only: that is the only version that exists when a new race comes along, and for target-aware steps (e.g. target encoding) the same mistake would leak a lot more.
- **E2, step order.** Scaling before imputing gives 0.7165 against 0.7166. The only difference is whether the imputed values are counted when the scaler computes its mean and standard deviation; that shifts the scale a little but not the order of the values, so a linear model barely notices.
- **E3, ingredients.** Without missing indicators validation PR-AUC moves +0.003 but ROC-AUC drops (0.9281 vs 0.9304); without scaling it moves +0.0015. Both are within noise for logistic regression. We keep the indicators because they let a model tell "no qualifying data / first race" apart from a real median value, and we keep scaling because the neural network trains with gradient descent and needs inputs on a similar scale.

Overall, preprocessing choices move validation PR-AUC by at most 0.003, much less than the choice of feature groups (Section 3) or model (Section 5).

**E4, training window.** Training on 2003-2019 gives the best average validation PR-AUC (0.7245, against 0.7181 for all seasons and 0.7062 for 1994+), so the code above keeps those seasons for every model from here on.

The gain is small (about 0.006, similar to seed-to-seed noise), so this is partly a judgement call. We follow the validation result because it agrees with the EDA: from 2003 every race has qualifying data, while the earlier decades (bigger fields, far worse reliability, no qualifying table) behave differently from the seasons we want to predict. The price is a smaller training set, 6,827 rows instead of 22,782. The feature evidence in Section 3 was computed on all training seasons, and the features themselves are defined the same way for every season.

## 5. Models and comparison

Every model below uses the same rows, the same features and the same preprocessing. Hyperparameters are chosen by **validation PR-AUC**, and the F1 threshold is the one that maximises F1 on **validation**; the test seasons are never used for either.

| Model | How it works | Main limitation here |
|---|---|---|
| Rule: grid <= 3 | no learning, predicts a podium for the top three starters | ignores car pace, form and reliability |
| Logistic regression (**baseline**) | a weighted sum of the features passed through a sigmoid; each weight is a change in log-odds | cannot model interactions (grid x team strength) or curved effects unless we build them by hand |
| Random forest | hundreds of decision trees on bootstrap samples; the probability is the average vote | many deep trees are hard to inspect; no extrapolation to unseen value ranges |
| Gradient boosting | small trees added one after another, each fitting the errors of the ones before | easy to overfit without early stopping / regularisation |
| Shallow neural network | 1-2 hidden ReLU layers trained with Adam, L2 penalty and early stopping | needs scaled, imputed inputs; less transparent; results move a little with the random seed |

The logistic regression with default settings is our **baseline**: the simplest reasonable model. Every other model has to beat it to justify its extra complexity.

**Primary metric: PR-AUC.** Only 13-15% of rows are podiums. PR-AUC measures how well the real podium finishers are ranked at the top of each race's field, ignores the many easy "back of the grid" negatives and needs no threshold. **Secondary: ROC-AUC**, the overall separation of the two classes (it looks high for every model because most negatives are easy). **F1, precision and recall** are reported at the threshold chosen on validation, applied unchanged to train and test.

**Train / validation / test** for every model (threshold for F1, precision and recall chosen on validation):

| model | ROC-AUC | PR-AUC | F1 | Precision | Recall |
|---|---|---|---|---|---|
| Rule: grid <= 3 | 0.909 / 0.909 / 0.914 | 0.626 / 0.639 / 0.647 | 0.646 / 0.692 / 0.649 | 0.647 / 0.692 / 0.650 | 0.646 / 0.692 / 0.647 |
| Logistic regression (baseline) | 0.933 / 0.932 / 0.940 | 0.698 / 0.725 / 0.720 | 0.667 / 0.704 / 0.674 | 0.573 / 0.669 / 0.585 | 0.799 / 0.744 / 0.794 |
| Logistic regression (tuned) | 0.933 / 0.931 / 0.939 | 0.692 / 0.727 / 0.716 | 0.666 / 0.699 / 0.669 | 0.571 / 0.659 / 0.581 | 0.800 / 0.744 / 0.789 |
| Random forest (tuned) | 0.955 / 0.932 / 0.939 | 0.776 / 0.721 / 0.707 | 0.719 / 0.714 / 0.691 | 0.663 / 0.702 / 0.648 | 0.785 / 0.726 / 0.740 |
| Gradient boosting (tuned) | 0.957 / 0.933 / 0.938 | 0.790 / 0.734 / 0.711 | 0.718 / 0.707 / 0.672 | 0.719 / 0.723 / 0.693 | 0.717 / 0.692 / 0.652 |
| Neural net (baseline, 1x32) | 0.933 / 0.923 / 0.933 | 0.693 / 0.728 / 0.686 | 0.668 / 0.715 / 0.654 | 0.587 / 0.674 / 0.585 | 0.776 / 0.761 / 0.740 |
| Neural net (tuned) | 0.938 / 0.926 / 0.932 | 0.715 / 0.731 / 0.713 | 0.675 / 0.704 / 0.648 | 0.592 / 0.662 / 0.571 | 0.785 / 0.752 / 0.750 |

**Reading the comparison.**

- **Primary model: gradient boosting**, the tuned model with the highest validation PR-AUC (0.734; neural net 0.731, logistic regression 0.727, random forest 0.721). On the unseen 2022-2024 seasons it reaches PR-AUC 0.711 and ROC-AUC 0.938. At the threshold chosen on validation (0.43) it predicts 192 podiums on the test seasons, 133 of them correct (precision 0.69), and finds 133 of the 204 real podiums (recall 0.65, F1 0.67).
- **Learning clearly beats the simple rule.** "Top three on the grid" already reaches test PR-AUC 0.647. Every learned model lands between 0.686 and 0.720, and ROC-AUC goes from 0.914 to about 0.94: recent form, team strength and standings tell the model which front-runners are likely to stay there and who can come from further back.
- **The extra complexity does not clearly pay off.** On the test seasons the plain logistic-regression baseline is actually the best (0.720), and every tuned model is within 0.013 of it. That is about the size of the seed-to-seed noise (+/-0.007 for the neural net) with only 68 test races. We keep gradient boosting as the primary model because that is what our validation rule picked, not because it is clearly better; the baseline would serve just as well here.
- **Different threshold trade-offs.** With its validation threshold, logistic regression catches more podiums (test recall 0.79) at lower precision (0.59), while gradient boosting is more conservative (0.65 / 0.69). The random forest has the best test F1 (0.69).
- **Overfitting check.** The two tree models fit the training seasons noticeably better than validation (PR-AUC 0.79 and 0.78 against 0.73 and 0.72). Logistic regression and the networks show no such gap; their training score is even a bit lower than validation, because the 2003-2013 part of the training data is harder to rank than recent seasons (PR-AUC by period: 0.766 for 2003-2013 against 0.828 for 2014-2019). The network stopped training after 10 epochs without improvement on its early-stopping split and kept the weights from its best epoch (dotted line in the training-curve plot).

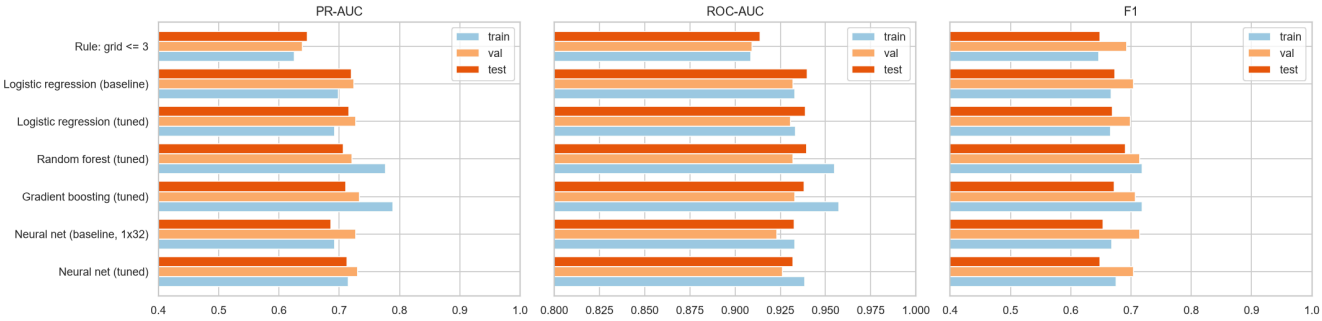



Figure 12. PR-AUC, ROC-AUC and F1 on train, validation and test.

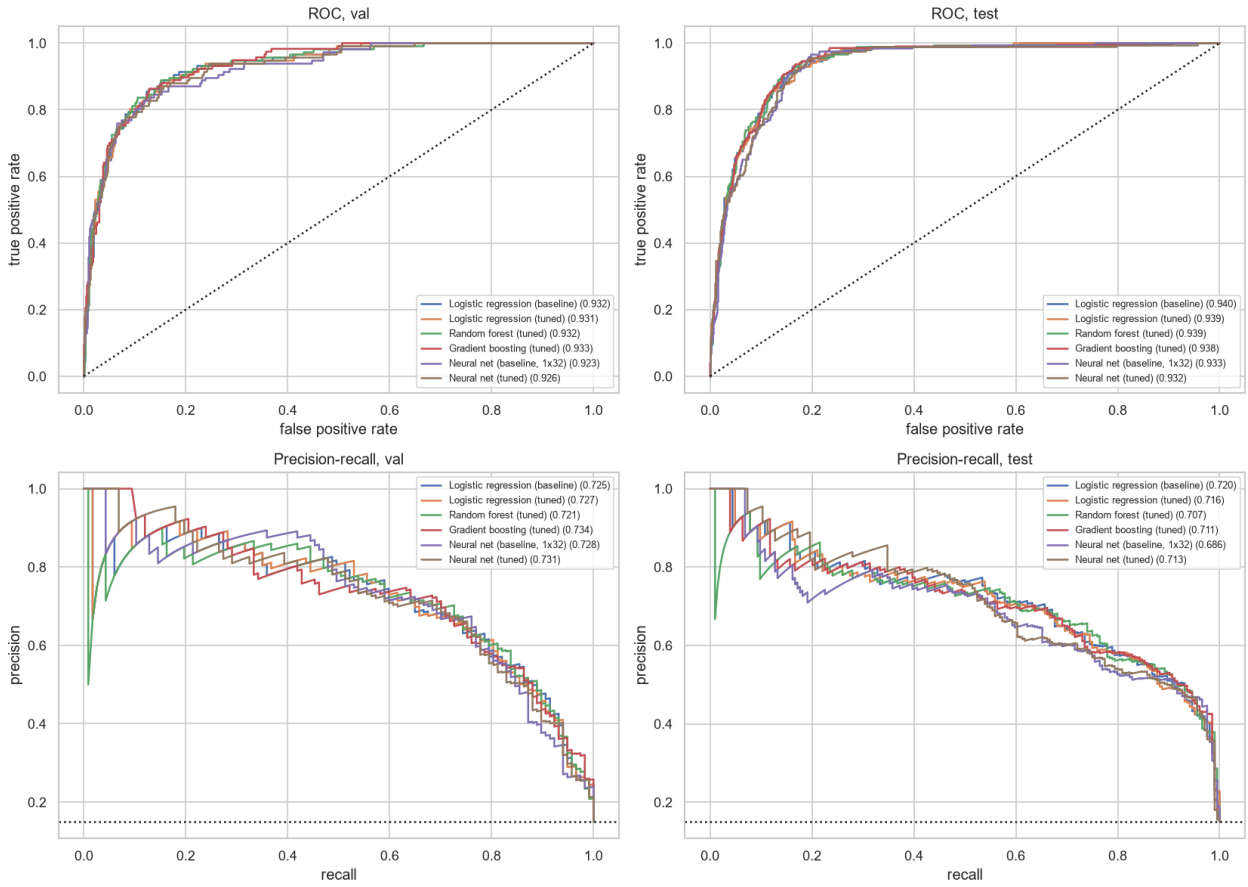



Figure 13. ROC and precision-recall curves on validation and test.

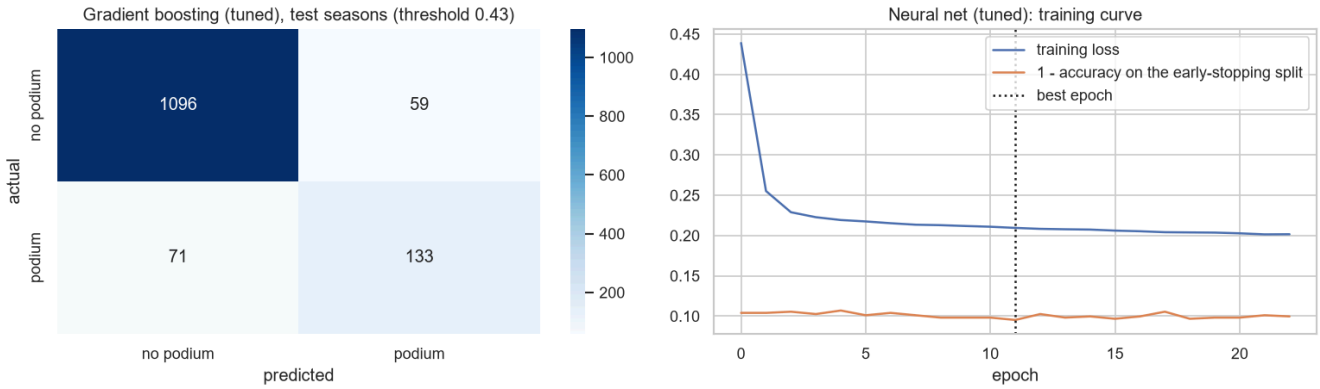



Figure 14. Confusion matrix of the primary model on the test seasons, and the neural network training curve.

## 6. Feature-group ablation

Section 3 justified the features *before* training. Here we measure what each group actually contributes *after* training: we remove one group at a time, retrain with the same tuned hyperparameters and compare PR-AUC with the full model. Because grid and qualifying carry overlapping information, we also remove the two together. To judge which changes are real, we refit the neural net with five different seeds.

| removed group | Logistic regression (val) | Gradient boosting (val) | Neural net (val) | Logistic regression (test) | Gradient boosting (test) | Neural net (test) |
|---|---|---|---|---|---|---|
| without grid | -0.045 | -0.035 | -0.068 | -0.074 | -0.047 | -0.090 |
| without quali | -0.009 | -0.002 | -0.022 | -0.004 | 0.013 | -0.012 |
| without driver_form | 0.001 | -0.016 | -0.032 | 0.012 | -0.013 | -0.007 |
| without constructor_form | -0.002 | 0.005 | -0.015 | -0.002 | -0.010 | -0.009 |

| | | | | | | |
|---|---|---|---|---|---|---|
| without standings | -0.014 | -0.027 | -0.025 | -0.006 | -0.009 | -0.005 |
| without context | 0.006 | -0.010 | -0.003 | 0.002 | -0.003 | -0.015 |
| without grid + quali | -0.090 | -0.076 | -0.093 | -0.147 | -0.148 | -0.163 |

**What the ablation shows.** (Seed noise for the neural net: test PR-AUC 0.697 +/- 0.007, so changes below about 0.015 are hard to tell apart from noise.)

- **Starting order is the one essential block.** Removing grid and qualifying together costs 0.076-0.093 validation PR-AUC and 0.147-0.163 test PR-AUC for all three models. Removing only the grid group already costs 0.035-0.068 on validation and 0.047-0.090 on test.
- **Qualifying on its own adds little once the grid is there** (validation -0.002 to -0.022, test between -0.012 and +0.013): the grid slot already contains most of what qualifying says.
- **Standings are a small but consistent help**: validation drops by 0.014-0.027 and test by 0.005-0.009 for every model when they are removed.
- **Driver form** matters for the two non-linear models on validation (-0.016 and -0.032) but not for logistic regression. **Constructor form and context** stay within about +/-0.015 with mixed signs; they are second-order once grid and standings are known.

This is the same picture as the feature-selection trials before training: grid first, standings second, the rest fine-tuning.

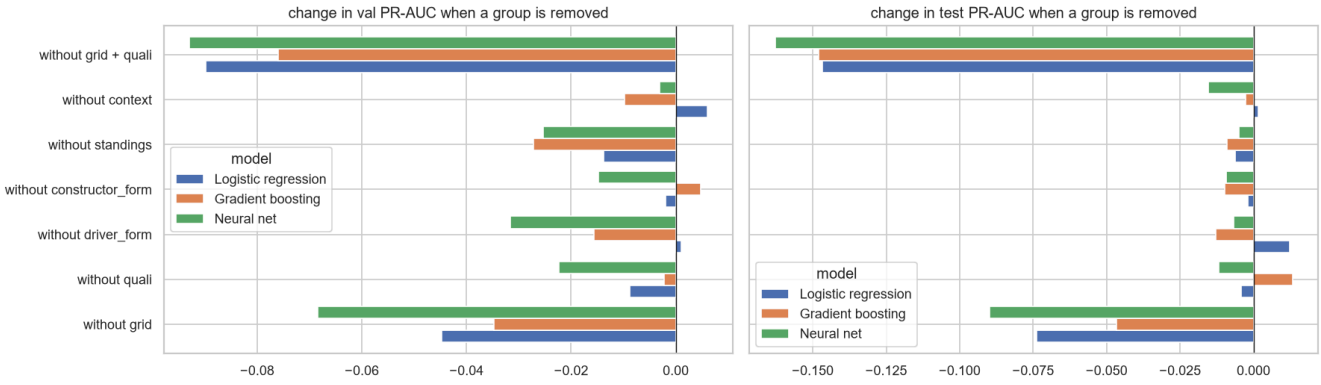



Figure 15. Change in PR-AUC when one feature group is removed.

## 7. Explainability

> **These explanations are not causal.** Permutation importance, SHAP and LIME describe what the fitted model relies on to make its predictions. They do not show what causes a podium: features stand in for things we cannot see (constructor form for car pace), they are correlated with each other, and the patterns only hold for the eras the model was trained on.

**Global explanations.**

- **The grid slot dominates.** For the gradient-boosting model, shuffling `grid_filled` costs 0.181 test PR-AUC, twice as much as the next three features together, and it has by far the largest mean |SHAP| (0.66 log-odds).
- **Then team strength and pace.** Both methods agree on the next three: the team's recent finishing positions (`con_finish_pct_5`, SHAP 0.40, permutation 0.046), the gap to pole (`quali_gap_pct`, 0.34 / 0.029) and the team's best grid slot (`team_best_grid`, 0.31 / 0.015). Constructor points share and the driver's podium rates follow in SHAP (0.13-0.14).
- **The directions make sense** (beeswarm): a low grid number, a small gap to pole, good recent team results and a high points share push the prediction up; a weak team record pushes it down strongly.

- **SHAP vs permutation.** Some standings and form features get SHAP weight but almost no permutation importance (e.g. `con_points_share`). These features are correlated with each other, so when one is shuffled the model still finds the information in the others. The ablation (Section 6) shows the same: removing one of these groups costs little.
- **The logistic regression** relies on the same top features in a slightly different mix: the grid slot first (0.140), then the team's best grid slot (0.044) and recent team results (0.036), and much less on the gap to pole (0.005).

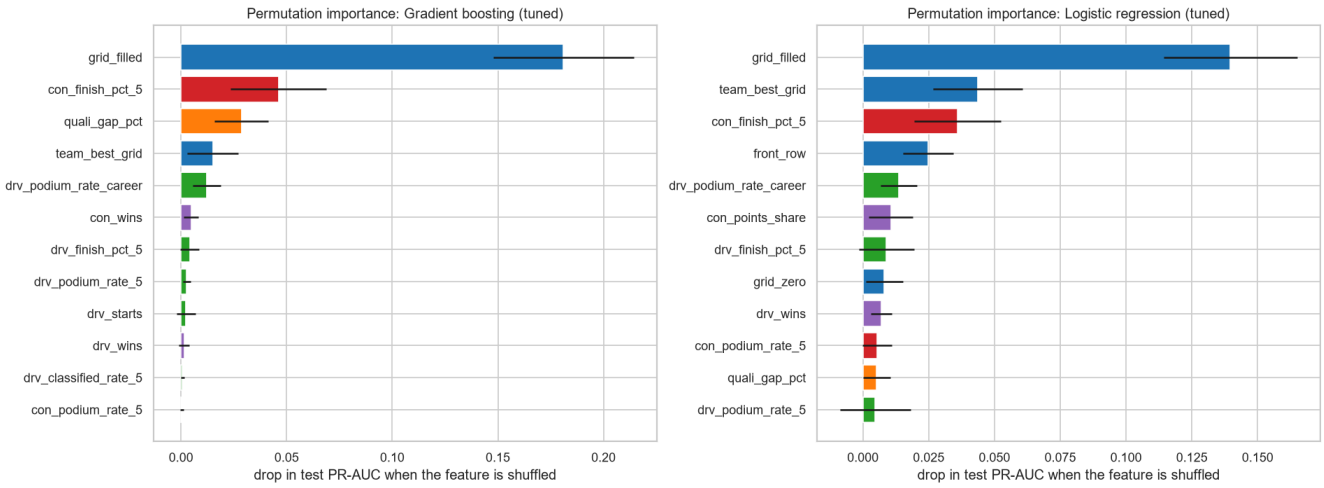



Figure 16. Permutation importance on the test seasons.

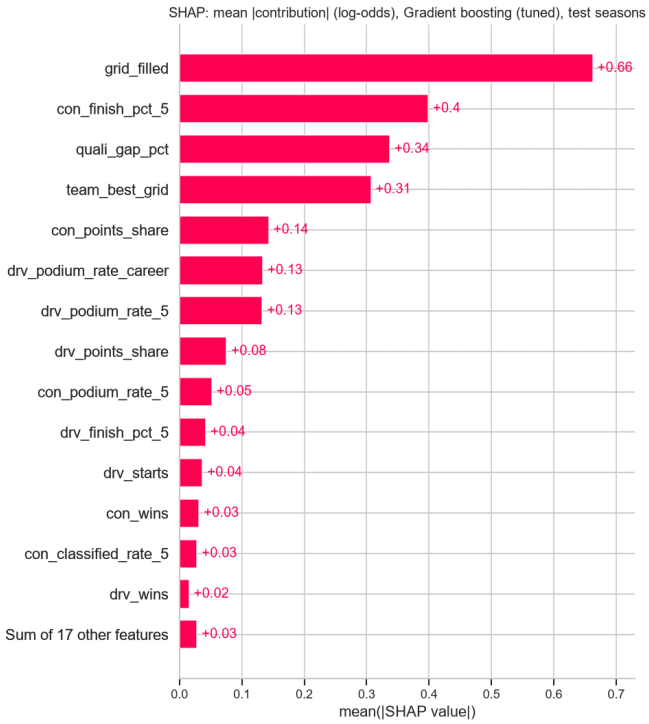



Figure 17. SHAP global importance.

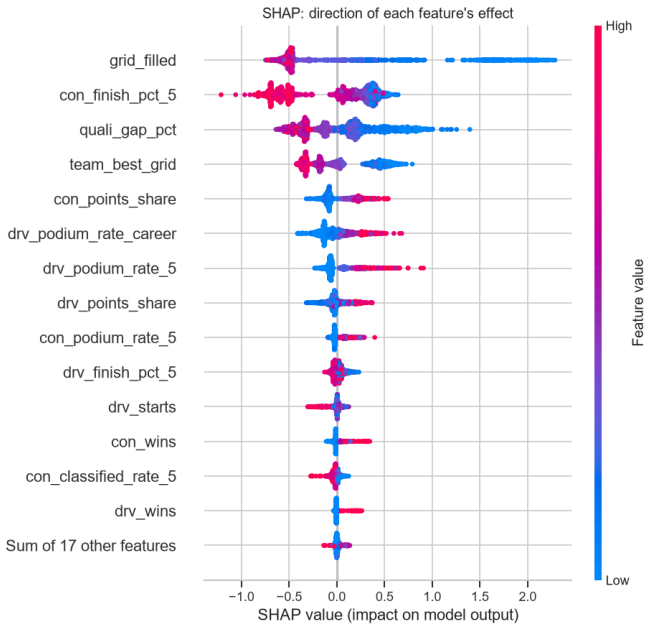



Figure 18. SHAP beeswarm.

**Local explanations.**

- **Typical podium: Charles Leclerc, Emilia Romagna GP 2024** (started P3, finished on the podium, predicted probability 0.675). SHAP starts from the model's average output (-3.20 log-odds) and the grid slot alone adds +1.64. A tiny gap to pole (0.3%) adds +0.60, the team's best grid slot (P3) +0.51, and good team form and points share add the rest. LIME's strongest rules say the same: `grid_filled <= 6`, `quali_gap_pct <= 0.88`, `con_finish_pct_5 <= 0.38` and a career podium rate above 0.20.

- **Surprise podium: Pierre Gasly, Dutch GP 2023** (started P12, finished P3, predicted probability 0.006). Every important feature pushes down: the team's weak recent results (`con_finish_pct_5` = 0.70, -0.59), the midfield grid slot (-0.48), the team's best slot (-0.33) and a 13.5% gap to pole (-0.30), which is unusually large because Gasly's best lap came from an earlier, slower session than the pole lap. LIME gives the same picture. The podium came from what happened in a rain-hit race, which is exactly the kind of information the model cannot have before the start.

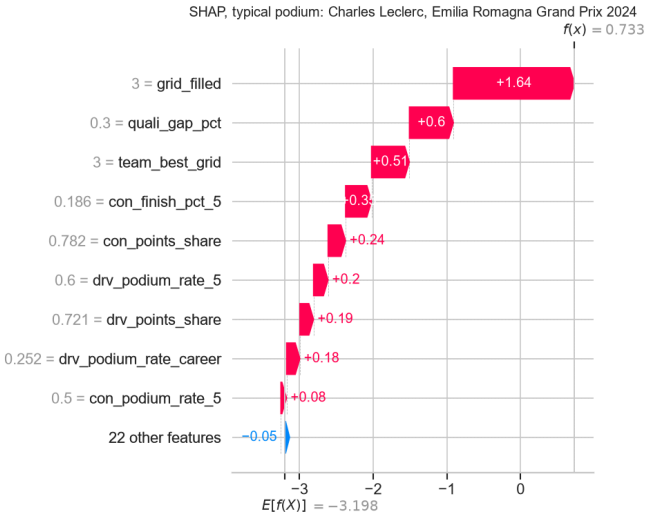



Figure 19. SHAP, typical podium.

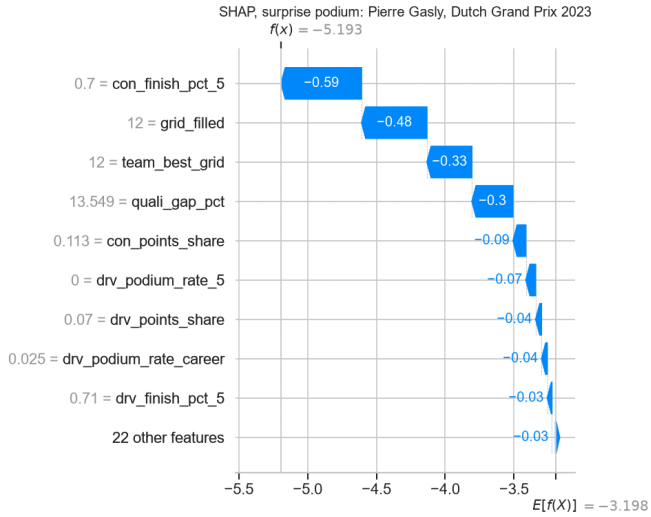



Figure 20. SHAP, surprise podium.

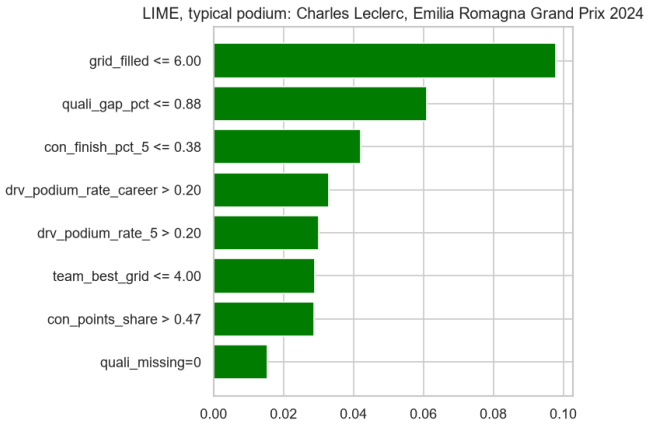



Figure 21. LIME, typical podium.

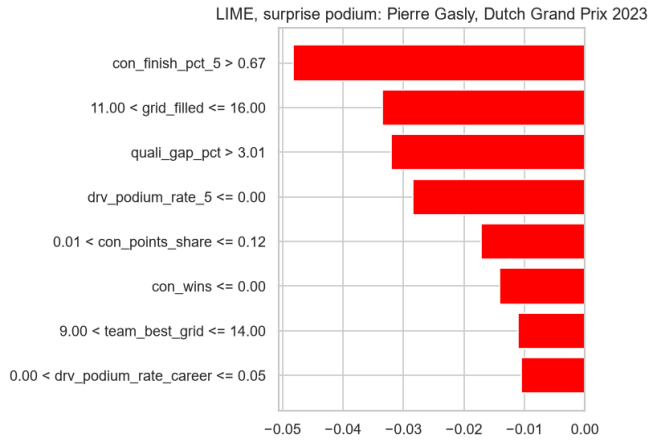



Figure 22. LIME, surprise podium.

## 8. Inference function

`predict_podium(record)` takes a dictionary with the pre-race features of one driver in one race. Any feature can be missing: it is imputed with the training median (plus its missing indicator), exactly as during training. It returns the podium probability, the yes/no decision at the validation-chosen threshold, and which features were missing.

We demonstrate it on a complete record (Charles Leclerc, Monaco 2024), on the same record with the qualifying and driver-form features removed, and on a record that only has a grid slot.

**Results.** With the complete record the model gives Leclerc a 0.813 podium probability (Leclerc won the race). Without the qualifying and driver-form features (8 values missing and imputed) the probability drops to 0.557, but the answer is still "podium". With only a grid slot of 3 and a field of 20 (19 values missing) it falls to 0.091: everything else is filled with training medians, which describe a midfield driver in a midfield car, so a P3 start alone is not enough to convince the model. The function never fails on missing data, but the less it is told, the closer its answer gets to "an average driver".

## 9. Known limitations

- **Qualifying-coverage drift.** `qualifying.csv` starts in 1994 and covers every race only from 2003, so 58% of the modelling rows have no qualifying data. Those values are imputed and flagged, and even where they exist they mean slightly different things across formats (single session in the 1990s, Q1/Q2/Q3 knockout since 2006).
- **Era sensitivity.** F1 changed a lot over 75 years, and older seasons are harder to rank (training PR-AUC 0.766 for 2003-2013 against 0.828 for 2014-2019). That is part of why validation picked a 2003-2019 training window, so the model has never seen racing before 2003. The test seasons are a new regulation era, and team-form features need a few races to catch up after a reset like 2022.
- **Small evaluation sets.** Validation has 39 races (117 podiums) and test 68 races (204 podiums). Differences of about 0.01-0.02 PR-AUC between models are within the noise we measured by changing the random seed.
- **Validation is used several times** (hyperparameters, threshold, training window), so validation scores are a little optimistic. The test seasons were only used for the final report.
- **Team renames** (Toro Rosso -> AlphaTauri -> RB, Renault -> Alpine, ...) get a new `constructorId`, which resets the constructor-form features.
- **Hand-made mappings**: nationality -> country (Q2), status -> mechanical failure (Q3), and the shared-drive rule. They are written out in full above, but someone else could reasonably draw the lines differently.
- **Missing information.** Weather, tyre strategy, safety cars, upgrades and first-lap incidents are not in the data. They explain most "surprise" podiums, like Gasly at Zandvoort 2023 in Section 7.
- **Cutoff edge case.** We treat the final grid as known before the start, including late pit-lane starts.
- **Explanations are associations, not causes** (Section 7).